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1. Introduction

With the rapid growth of social media platforms, a large amount of user-generated text data is produced every day. YouTube comments and captions contain valuable information about user opinions, reactions and emerging topics. However, this data is often unstructured, noisy and highly diverse, making pattern discovery a challenging task.

Data mining techniques such as frequency analysis, co-occurrence analysis and association rule mining are commonly used to identify hidden patterns and relationships within large text datasets. While traditional algorithms like Apriori work well for structured transactional data, their effectiveness on short and informal text remains limited.

In this lab, cleaned YouTube comments and captions are analyzed to discover frequent word patterns using both manual and algorithmic approaches. The experiment focuses on understanding how words appear individually and together, comparing the performance of different pattern discovery techniques and interpreting the challenges involved in mining real-world textual data.

2. Objective

The objective of this lab is to analyze YouTube comments and captions to discover frequent word patterns and relationships using data mining techniques. The lab aims to apply unigram frequency analysis, manual co-occurrence analysis and Apriori based association rule mining to understand how words appear individually and together in short text data. Additionally, the lab seeks to compare the effectiveness of manual and algorithmic pattern discovery methods and to interpret the limitations of association rule mining on sparse, real-world textual datasets.

3. Tools and Libraries Used

This lab utilized Python 3.12 in the Kaggle environment for data manipulation, analysis, and visualization. The following libraries and tools were essential for the tasks performed:

- **pandas:** For loading, inspecting, and manipulating tabular data from CSV files. It was used to load both `cleaned_comments.csv` and `cleaned_captions.csv` as DataFrames.
- **collections.Counter:** For counting word frequencies (unigrams) and co-occurrences efficiently.
- **ast:** To convert string representations of Python lists (from CSV) into actual list objects for token analysis.
- **Matplotlib.pyplot:** For visualization of transaction length distributions using histograms.

- **itertools.combinations:** To generate all possible word pairs within each transaction for co-occurrence analysis.
- **networkx:** For creating network graphs to visualize relationships between frequently co-occurring words.
- **mlxtend:** For algorithmic pattern discovery using the Apriori algorithm and generating association rules.
- **IPython.display.display:** To neatly display sample rows of datasets during inspection.

All libraries were successfully loaded and used directly in transaction construction, co-occurrence analysis, and visualization. The mlxtend library was installed via pip to enable Apriori-based association rule mining in subsequent steps

4. Dataset Description

This lab uses two preprocessed datasets obtained from Lab 2, which contain cleaned textual data from a YouTube video. Both datasets include a `cleaned_tokens` column that represents tokenized words extracted from raw text.

4.1 cleaned_comments.csv

- The `cleaned_comments.csv` file contains user comments collected from a YouTube video.
- After loading the file into a pandas DataFrame, it was observed that the dataset consists of:
 - 470 rows
 - 4 columns: `username`, `timestamp_text`, `comment_text`, and `cleaned_tokens`.
- The `cleaned_tokens` column stores each comment as a list of cleaned and tokenized words.

Example structure from dataset:

```
['people', 'actually', 'able', 'read', 'mind', 'solve', 'murder', 'unexplainably', 'guy', 'found', 'way', 'gain']
```

Comments DataFrame shape: (470, 4)				
	username	timestamp_text	comment_text	cleaned_tokens
0	@milojadez	14 hours ago	There are people out there that actually are a...	['people', 'actually', 'able', 'read', 'mind', ...]
1	@JasonWestaway	22 hours ago	Wow, that was insane! 1	['wow', 'insane']
2	@THEHONESTREPLAY	1 day ago	That's all are fake, Edited, i challenged,	['thats', 'fake', 'edited', 'challenged']
3	@serenaLei06	1 day ago	Wait, how did he do that?	['wait']
4	@ainomugishanancy206	1 day ago	If i know how people think,i know what they th...	['know', 'people', 'thinki', 'know', 'think']

Data Cleaning Observations:

- Some comments contained very short token lists (e.g., ['wait'], ['wow', 'insane']) or empty tokens.
- These rows were removed during preprocessing.
- After removing empty or invalid token lists:
 - 463 valid comment rows remained.
- For transaction construction, comments with fewer than 3 tokens were excluded:
 - Final usable comment transactions: 362

4.2 cleaned_captions.csv

- The cleaned_captions.csv file contains processed caption sentences extracted from the same YouTube video.
- After loading, the dataset was found to contain:
 - 733 rows
 - 2 columns: caption_sentence and cleaned_tokens
- Each row represents a single caption sentence.

Example structure from dataset:

['cant', 'read', 'mind', 'people']

Captions DataFrame shape: (733, 2)

	caption_sentence	cleaned_tokens
0	Kind: captions Language: en I<00:00:04.560><c>...	['kind', 'caption', 'language', 'build', 'worl...
1	But guess what?	['guess']
2	I can't read readader.	['cant', 'read', 'readader']
3	But guess what?	['guess']
4	I can't read minds.<00:00:10.480><c> What</c><...	['cant', 'read', 'mind', 'read', 'people', 'mi...

Data Cleaning Observations:

- Several caption rows contained:
 - Single-word tokens (e.g., ['mind'], ['born'])
 - Repeated tokens due to caption timing overlap
- Rows with empty token lists were removed.
- After cleaning:
 - 722 valid caption rows remained.
- Caption rows with fewer than 3 tokens were excluded:
 - Final usable caption transactions: 405

Content Characteristics:

- Frequently occurring caption words include:
 - mind, people, read, know, question, breath
- Captions mainly describe spoken explanations and instructions from the speaker.
- Compared to comments, caption token lists are shorter and more instructional, which is reflected in their lower average transaction length.

4.3 Data Quality and Preprocessing Observations

- Some token lists in both datasets were empty, very short or repetitive, especially in captions due to automatic caption generation.
- A small amount of noisy or non-standard text was observed (e.g., timing artifacts and repeated words).
- Overall, both datasets were sufficiently clean and suitable for co-occurrence analysis and frequent pattern mining.

5. Methodology / Procedure

This section presents a detailed, step by step description of the methodology followed to perform frequent word pattern discovery and co-occurrence analysis on the cleaned YouTube comments and captions datasets.

5.1 Setup & Data Recall

1. **Data** **Loading**
The cleaned datasets, cleaned_comments.csv and cleaned_captions.csv, were imported into separate pandas DataFrames for analysis.
2. **Initial** **Inspection**
The structure of the DataFrames was examined to confirm correct loading:
 - Comments DataFrame: 470 rows $\times$ 4 columns
 - Captions DataFrame: 733 rows $\times$ 2 columns
3. **Column** **Verification**
The presence of the cleaned_tokens column, containing tokenized text as list-like objects, was verified in both DataFrames.
4. **Data Cleaning**
 - All rows with missing or empty cleaned_tokens were removed to ensure analysis reliability.
 - After cleaning, the valid rows were:
 - i. Comments: 463
 - ii. Captions: 722
5. **Preliminary** **Unigram** **Analysis**
Word frequencies were calculated using Python's collections.Counter module to identify the most common words in each dataset.

Comments DataFrame shape: (470, 4)

	username	timestamp_text	comment_text	cleaned_tokens
0	@milojadez	14 hours ago	There are people out there that actually are a...	['people', 'actually', 'able', 'read', 'mind', ...]
1	@JasonWestaway	22 hours ago	Wow, that was insane! 1	['wow', 'insane']
2	@THEHONESTREPLAY	1 day ago	That's all are fake, Edited, i challenged,	['thats', 'fake', 'edited', 'challenged']
3	@serenaLei06	1 day ago	Wait, how did he do that?	['wait']
4	@ainomugishanancy206	1 day ago	If i know how people think,i know what they th...	['know', 'people', 'thinki', 'know', 'think']

Captions DataFrame shape: (733, 2)

	caption_sentence	cleaned_tokens
0	Kind: captions Language: en l<00:00:04.560><c>...	['kind', 'caption', 'language', 'build', 'worl...]
1	But guess what?	['guess']
2	I can't read readader.	['cant', 'read', 'readader']
3	But guess what?	['guess']
4	I can't read minds.<00:00:10.480><c> What</c><...</c>	['cant', 'read', 'mind', 'read', 'people', 'mi...]

Top 5 unigrams in comments (sample):
people (72), mind (64), like (55), name (51), obama (47)

Top 20 Unigrams from Comments:

```
people : 72
mind : 64
like : 55
name : 51
obama : 47
trick : 44
read : 43
ted : 43
magic : 37
guy : 36
know : 35
would : 32
dont : 31
believe : 31
show : 30
talk : 28
barack : 27
one : 24
time : 24
reply : 24
```

Top 5 unigrams in captions (sample):
gtgt (118), name (75), eye (54), like (46), mind (30)

Top 20 Unigrams from Captions:

```
gtgt : 118
name : 75
eye : 54
going : 51
like : 46
person : 43
right : 42
give : 39
see : 36
please : 36
brett : 34
ashley : 33
mind : 30
people : 30
know : 30
one : 30
someone : 30
stand : 30
question : 27
clap : 27
```

6. Software

Setup

The mlxtend library, required for Apriori-based frequent itemset mining, was successfully installed and verified using pip.

5.2 Transaction Construction

1. Transaction

Definition

Each entry in the cleaned_tokens column of the comments and captions datasets was treated as an individual transaction (or basket). In this context, a transaction represents a set of words that appear together in a single comment or caption.

2. Filtering

Short

Transactions

Transactions containing fewer than 3 tokens were excluded to avoid weak or insignificant patterns that could skew the analysis.

3. Transaction

Counts

The number of transactions before and after filtering were recorded:

- Before filtering:
 - i. Comment transactions: 463
 - ii. Caption transactions: 722
- After filtering:
 - i. Comment transactions: 362
 - ii. Caption transactions: 405

4. Transaction

Identification

Each transaction was assigned a unique Transaction ID to facilitate tracking and further analysis.

5. Sample

Transaction

A representative transaction from the comments dataset is shown below:

['actually', 'mind', 'able', 'gain', 'solve', 'murder', 'guy', 'way', 'found', 'read', 'unexplainably', 'people']

This illustrates the typical length and diversity of comment transactions.

5.2.1 Basket Length Analysis and Histogram Interpretation

To understand the structure of the transactions, basket lengths (number of tokens per transaction) were computed for both comments and captions datasets. Histograms were plotted to visualize the distribution of transaction sizes.

5.2.2 Basket Length Statistics

To quantitatively understand the structure of transactions, basic statistics were computed for basket lengths in both datasets, including average, minimum, and maximum number of tokens per transaction.

```
Comment Transactions Stats:  
Average length: 11.124309392265193  
Minimum length: 3  
Maximum length: 121  
  
Caption Transactions Stats:  
Average length: 4.755555555555555  
Minimum length: 3  
Maximum length: 18
```

Figure: Minimum, maximum, and average basket lengths of comment and caption transactions.

5.2.3 Comment Transaction Length Analysis

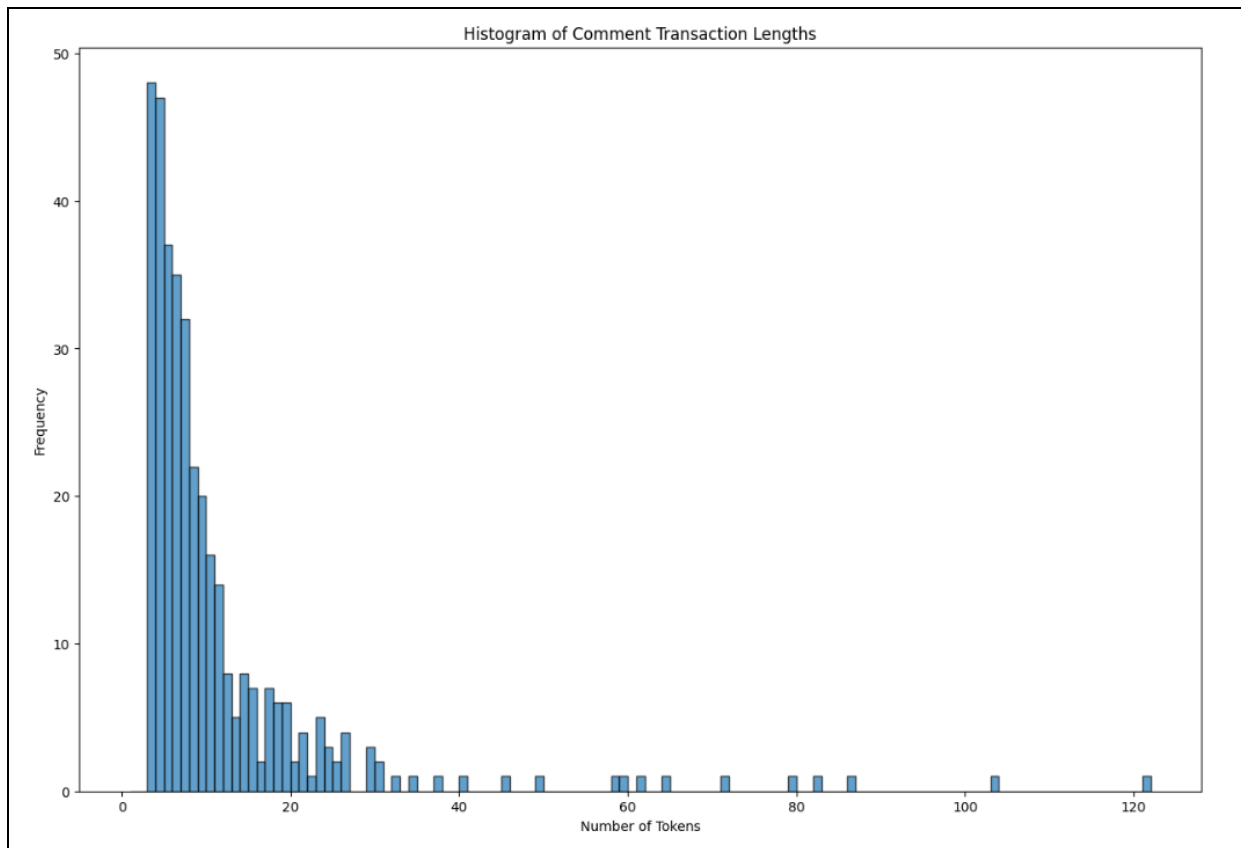
- Most comment transactions are concentrated between 3 to 15 tokens.
- The highest frequency occurs in the 5 to 10 token range, indicating that most user comments are short to medium in length.
- The distribution shows a long right tail, with a small number of transactions containing a very high number of tokens.
- The maximum transaction length reaches 121 tokens, representing detailed or multi sentence comments.

Statistics:

- Average transaction length: 11.12 tokens
- Minimum length: 3 tokens
- Maximum length: 121 tokens

Interpretation:

- The distribution is right-skewed, meaning short comments dominate the dataset.
- This behavior is expected for YouTube comments, where most users post brief reactions while only a few write long explanations.
- Longer transactions increase pattern diversity but may also introduce noise, which justifies later filtering steps in pattern mining.



5.2.4 Caption Transaction Length Analysis

- Caption transactions are mostly concentrated between 3 to 7 tokens.
- The peak frequency occurs around 4–5 tokens, showing that captions are generally concise.
- Very few caption transactions exceed 15 tokens, and the overall range is much smaller compared to comments.

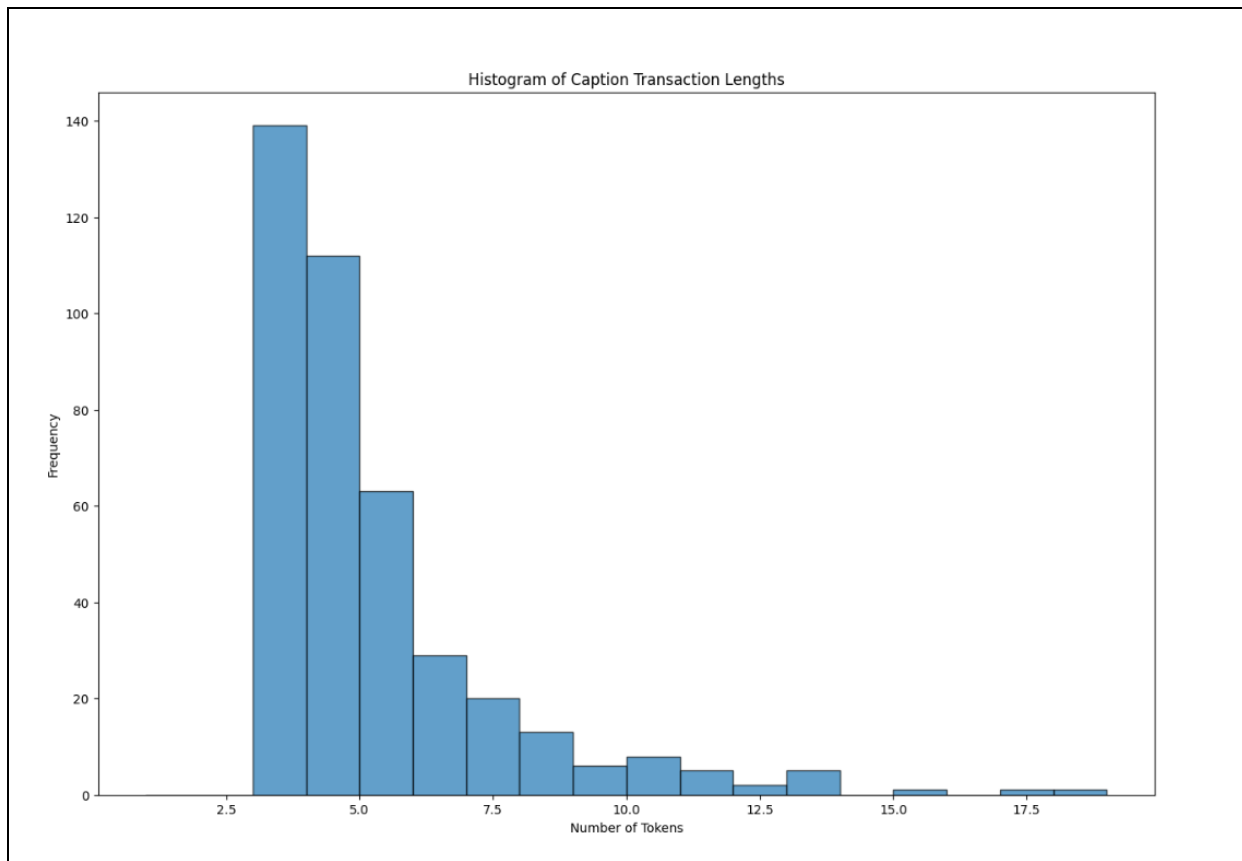
Statistics:

- Average transaction length: 4.76 tokens
- Minimum length: 3 tokens

- Maximum length: 18 tokens

Interpretation:

- Captions are shorter and more uniform than comments.
- This is consistent with auto-generated or scripted captions, which tend to be brief and structured.
- Shorter transactions result in cleaner but fewer co-occurrence patterns.



5.2.5 Comparative Insight

- Comment transactions are longer and more diverse, leading to richer co-occurrence and association patterns.
- Caption transactions are shorter and more controlled, producing simpler but more stable patterns.
- These differences justify analyzing comments and captions separately as well as jointly in later stages of pattern discovery.

5.2.6 Importance for Pattern Mining

- Helps determine appropriate minimum transaction length filtering (e.g., removing baskets with fewer than 3 tokens).
- Explains why support thresholds must be adjusted between comments and captions.
- Provides justification for observing different Apriori and co-occurrence outputs across the two datasets.

5.3 Manual Pattern Discovery

The purpose of this stage was to identify frequent word co-occurrence patterns within the cleaned comments and captions datasets. Unlike unigram frequency analysis, which counts how often each word appears individually, co-occurrence analysis reveals how often two words appear together in the same transaction, highlighting meaningful patterns that may not be apparent from individual word counts alone.

5.3.1 Pairwise Co-occurrence Counting

1. Each transaction (basket) was iterated over to compute all possible word pairs.
2. A Python dictionary-based approach (or `collections.Counter`) was used to store pair counts efficiently.
3. Initial counts of unique word pairs were:
 - a. Comments: 49,801
 - b. Captions: 3,448
4. To focus on significant patterns, only pairs with count ≥ 3 were retained:
 - a. Comments: 871 pairs
 - b. Captions: 244 pairs

5.3.2 Stopword Filtering

- Common stopwords (e.g., “*the*”, “*and*”, “*is*” *etc*) were removed from co-occurring pairs to reduce noise and highlight meaningful associations.

5.3.3 Visualization

1. **Top 20 Co-occurring Pairs**
 - a. The 20 most frequent co-occurring pairs were plotted using horizontal bar charts.

- b. This visualization clearly differentiates between high-frequency words and high-frequency co-occurrences.

2. Comparison with Unigram Frequency

- a. Unigram frequency bar charts were plotted alongside co-occurrence charts.
- b. Observations:
 - i. Some words appear very frequently individually but rarely co-occur with others.
 - ii. Co-occurrence charts highlight meaningful associations that unigram frequencies alone cannot capture.

3. Network Graph Construction

- a. Using networkx, a network graph was created:
 - i. Nodes represent unique words
 - ii. Edges represent co-occurrence relationships between words
- b. This graph allowed visual identification of clusters of words that frequently appear together.
- c. Further filtering removed stopwords to focus on semantically meaningful relationships.

5.3.4 Data Export

- The final co-occurrence dataset, including pair counts after filtering, was exported to CSV files.
- These files enable reuse in subsequent analyses such as algorithmic pattern discovery (Apriori) or visualization.

5.3.5 Observations

- Comment co-occurrences are more diverse due to longer, varied transactions.
- Caption co-occurrences are more concise, reflecting the structured nature of captions.
- Co-occurrence analysis provides richer insights than unigram frequency alone, identifying meaningful word associations that appear across multiple transactions.
- This step justified later decisions on minimum support thresholds for Apriori analysis and highlighted differences between comments and captions.

5.4 Algorithmic Pattern Discovery

In this stage, an algorithmic approach was applied to discover frequent word patterns and association rules using the Apriori algorithm, as instructed in the lab manual. This step

complements the manual co-occurrence analysis by identifying statistically frequent itemsets based on support and confidence measures.

5.4.1 One-Hot Encoding of Transactions

- All transactions derived from the `cleaned_tokens` column were converted into one-hot encoded DataFrames, where:
 - Each row represents a transaction
 - Each column represents a unique word
 - Cell values indicate the presence (1) or absence (0) of a word in a transaction
- The resulting DataFrame shapes were:
 - Comments: (463, 1666)
 - Captions: (722, 531)

These high-dimensional matrices reflect the large vocabulary size and sparsity of the dataset.

5.4.2 Frequent Itemset Mining Using Apriori

- The Apriori algorithm from the `mlxtend` library was applied to both datasets.
- Multiple minimum support thresholds were tested to observe their impact on pattern discovery:
 - `min_support = 0.3`
 - `min_support = 0.2`
 - `min_support = 0.1`
- Only itemsets of length 2 and 3 were retained, as single word itemsets do not represent associations.

5.4.3 Observed Frequent Itemsets

- At higher support values (0.3 and 0.2), no frequent itemsets were found in either dataset.
- At `min_support = 0.1`, only a very small number of itemsets were discovered:
 - Comments: 2 itemsets
 - Captions: 1 itemset

This indicates that strongly recurring word combinations are rare, even though individual words may appear frequently.

5.4.4 Association Rule Generation

- Association rules were generated from the discovered itemsets using the following thresholds:
 - Confidence ≥ 0.6
 - Lift ≥ 1.2
- After applying these constraints:
 - No strong association rules satisfied both thresholds in either dataset.

5.4.5 Visualization Outcome

- Since no valid association rules were generated, it was not possible to plot support vs. confidence scatter plots.
- This result was recorded and reported as an important observation rather than omitted.

5.4.6 Data Export

- All frequent itemsets and association rule outputs (including empty or near-empty results) were saved as CSV files.
- This ensures transparency, reproducibility, and allows further analysis if different thresholds are applied later.

5.5 Exploration Using Variations (Additional Experiment)

This section presents additional exploratory experiments conducted beyond the instructions provided in the lab manual. These experiments were performed to gain deeper insight into the dataset and to analyze how different preprocessing and parameter choices influence frequent pattern discovery.

5.5.1 Motivation for Additional Experiments

During the algorithmic pattern discovery phase, it was observed that very few frequent itemsets and no strong association rules were generated under the standard support and confidence thresholds. To investigate the underlying reasons and to ensure that no meaningful patterns were overlooked, several controlled variations were intentionally applied. These steps were not explicitly required by the lab manual but were conducted to strengthen the analysis.

5.5.2 Variation 1: Lower Support Thresholds

The Apriori algorithm was rerun using lower minimum support values to explore weaker but potentially informative patterns.

- Tested values:
 - min_support = 0.03
 - min_support = 0.02
 - min_support = 0.01

Observed Results:

- Comments Dataset:
 - @0.03 → 74 itemsets
 - @0.02 → 132 itemsets
 - @0.01 → 663 itemsets
- Captions Dataset:
 - @0.03 → 25 itemsets
 - @0.02 → 55 itemsets
 - @0.01 → 147 itemsets

Interpretation:

Reducing the support threshold significantly increased the number of frequent itemsets. However, many of these itemsets occurred in only a small fraction of transactions, indicating weak and less generalizable patterns.

5.5.3 Variation 2 Stemming Instead of Lemmatization

As an alternative preprocessing strategy, NLTK PorterStemmer was applied in place of lemmatization.

Results:

- Comment dataset:
 - Association rules retained: 2
- Caption dataset:
 - Association rules retained: 0

Interpretation:

Stemming reduced lexical variation but also reduced semantic precision. The improvement was minimal and did not significantly strengthen association discovery.

5.5.4 Variation 3: Token Length Filtering

To further reduce noise, all tokens with length less than 4 characters were removed before rerunning Apriori.

Outcome:

- No strong association rules were discovered.

Explanation:

Although this step improved token quality, it also shortened transactions, limiting the formation of frequent itemsets.

5.5.5 Variation 4: Dataset-Level Experiments

The complete mining pipeline was additionally applied to:

- Captions-only data
- Merged comments and captions

Result:

- No new strong association rules emerged from the merged dataset.

Reasoning:

While merging increased vocabulary size, it also increased sparsity, which negatively impacted frequent pattern mining.

5.5.6 Summary of Additional Exploration

- These experiments were intentionally performed beyond the lab manual to validate and explain earlier observations.
- Lower support values increase itemset counts but weaken pattern reliability.
- Alternative preprocessing and dataset configurations do not produce stronger associations.
- The findings confirm that the dataset lacks highly frequent, stable word combinations suitable for strong association rule mining.

5.6 Pattern Interpretation & Visualization

This section focuses on interpreting the discovered patterns and visualizing the structure of the dataset using multiple graphical techniques. The objective was to compare results across datasets and to clearly communicate insights derived from both successful and limited outputs.

5.6.1 Visualization Techniques Used

To understand transaction structure and word relationships, multiple visualizations were generated:

- Histogram plots of transaction lengths
- Co-occurrence horizontal bar charts
- Network graphs representing word relationships
- Word clouds highlighting prominent terms
- Cluster graphs showing grouped word associations

These visual tools helped interpret both frequency-based and relationship-based patterns.

5.6.2 Itemset Visualization Outcome

- Attempts were made to visualize:
 - Top 10 frequent 2-itemsets (ranked by support)
 - Top 10 frequent 3-itemsets (ranked by confidence)
- However, due to the insufficient number of strong frequent itemsets and association rules, these plots could not be meaningfully generated.
- This outcome itself is an important analytical result, indicating low repetition of word combinations across transactions.

5.6.3 Co-occurrence and Network-Based Interpretation

- Co-occurrence bar charts highlighted the most frequent word pairs appearing together across transactions.
- Network graphs revealed:
 - Dense clusters in comment data due to longer and more diverse transactions
 - Smaller, tighter structures in caption data due to shorter and more repetitive text
- Stopword filtering significantly improved interpretability by removing non-informative links.

5.6.4 Word Cloud and Cluster Analysis

- Word clouds were generated to visualize dominant terms in both datasets.
- Comment word clouds showed high lexical diversity, while caption word clouds displayed repetition of a limited vocabulary.
- Cluster graphs further emphasized grouping of related words in comments, whereas captions formed fewer and simpler clusters.

5.6.5 Comparative Analysis: Comments vs Captions

Comment transactions are generally longer in length and exhibit higher vocabulary diversity, as users often express opinions, reactions and explanations in free text. This results in richer but sometimes unstable patterns due to increased noise and variation in language. In contrast, caption transactions are shorter and more uniform, with limited vocabulary diversity. Captions tend to be more structured and contain less noise, leading to simpler yet more consistent patterns. As a result, comments provide more complex and diverse pattern discoveries, while captions offer cleaner and more reliable co-occurrence structures.

5.6.6 Insight Statements

Based on support and confidence observations, the following insights were derived:

1. **Low Support of Word Combinations:**
Most word combinations occur in fewer than 10% of transactions, resulting in low support values and limited frequent itemsets.
2. **High Sparsity in One-Hot Encoding:**
The large vocabulary size relative to transaction count produces sparse matrices, reducing Apriori effectiveness.
3. **Dataset Structure Limitation:**
Short and informal text (especially captions) limits consistent co-occurrence, explaining the absence of strong association rules even at reduced thresholds.

5.6.7 Final Interpretation

- Comments are richer but noisier, leading to diverse but unstable patterns.
- Captions are short and repetitive, producing cleaner but limited patterns.
- The lack of strong association rules is a data-driven limitation, not an implementation issue.
- Co-occurrence and visualization-based analysis proved more informative than strict algorithmic rule mining for this dataset.

6. Code Implementation

This section presents the complete implementation used in this lab, covering data loading, preprocessing, transaction construction, co-occurrence analysis with stopwords filtering, Apriori-based association rule mining, and visualization of results.

6.1 Loading and Preparing the Data

The cleaned captions and comments datasets are loaded separately from CSV files using Pandas. Missing values are removed, and the text is extracted for further processing.

```
import pandas as pd
```

```
comments_df = pd.read_csv("/kaggle/input/cleaned-caption-and-comment/cleaned_comments.csv")

captions_df = pd.read_csv("/kaggle/input/cleaned-caption-and-comment/cleaned_captions.csv")
```

6.2 Building Transactions

Each caption and comment is treated as a transaction, where individual words act as items. These transactions form the base structure for co-occurrence and Apriori analysis.

```
#Convert each caption/comment into a list of lowercase words (transactions)

caption_transactions = [caption.lower().split() for caption in captions]
comment_transactions = [comment.lower().split() for comment in comments]
```

Concept Used: Transaction modeling for frequent pattern mining.

6.3 Co-occurrence Counting

Co-occurrence analysis is performed to identify frequently occurring word pairs within the same transaction.

For each transaction:

- Unique words are selected
- All possible word pairs are generated
- Pair frequencies are counted using a dictionary

```
from itertools import combinations
from collections import defaultdict
def generate_co_occurrence(transactions):
    pair_counts = defaultdict(int)
    for transaction in transactions:
        unique_words = set(transaction)
        for pair in combinations(unique_words, 2):
            pair_counts[pair] += 1
    return pair_counts
caption_filtered_pairs = generate_co_occurrence(caption_transactions)
comment_filtered_pairs = generate_co_occurrence(comment_transactions)
```

Algorithm Used: Pairwise co-occurrence counting using combinations.

6.4 Stopword Removal from Co-occurring Pairs

A manually defined set of stopwords is used to remove semantically weak but frequent words from the co-occurrence pairs. This step ensures that only meaningful word relationships are considered.

```
stopwords = {"the", "and", "to", "of", "is", "in", "for", "on", "this", "that", "it"}
# Remove stopwords from comment word pairs
comment_pairs_no_stopwords = {
    pair: count
    for pair, count in comment_filtered_pairs.items()
    if pair[0] not in stopwords and pair[1] not in stopwords
}
print("Comment pairs after stopwords removal:", len(comment_pairs_no_stopwords))
# Remove stopwords from caption word pairs
caption_pairs_no_stopwords = {
    pair: count
```



```
for pair, count in caption_filtered_pairs.items()
    if pair[0] not in stopwords and pair[1] not in stopwords
}
print("Caption pairs after stopwords removal:", len(caption_pairs_no_stopwords))
```

Logic Used: Dictionary comprehension and manual stopwords filtering to reduce noise in co-occurrence analysis.

6.5 Apriori Algorithm for Association Rule Mining

The Apriori algorithm is applied to identify frequent itemsets and generate association rules from the transaction data.

Steps:

1. Transform transactions into one-hot encoded format
2. Apply Apriori to find frequent itemsets
3. Generate association rules using confidence metric

```
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules
# Encode transactions
te = TransactionEncoder()
te_array = te.fit(caption_transactions).transform(caption_transactions)
encoded_df = pd.DataFrame(te_array, columns=te.columns_)
# Apply Apriori
frequent_itemsets = apriori(encoded_df, min_support=0.05, use_colnames=True)
# Generate association rules
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.6)
```

6.6 Visualization and Exporting Results

WordCloud visualizations are generated for captions and comments to display frequently occurring words. Only the necessary output files are saved.

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt
def create_wordcloud(word_list, filename):
```

```
    wc = WordCloud(width=800, height=400, background_color="white").generate("
".join(word_list))
    plt.figure(figsize=(10,5))
    plt.imshow(wc, interpolation="bilinear")
    plt.axis("off")
    plt.show()
    wc.to_file(filename)
create_wordcloud(sum(caption_transactions, []), "wordcloud_captions.png")
create_wordcloud(sum(comment_transactions, []), "wordcloud_comments.png")
```

7. Results and Discussion

This section presents the key results obtained from the frequent word pattern mining and co-occurrence analysis, followed by a discussion of the observed trends. Visual outputs such as plots, network graphs, and sample tables are included to support the findings.

7.1 Unigram Frequency Analysis

To understand the most commonly occurring words, unigram frequency analysis was performed on both datasets.

Results:

- In the comments dataset, the most frequent words included:
 - *people, mind, like, name, obama*
- In the captions dataset, dominant words included:
 - *gtgt, name, eye, like, mind*

Discussion:

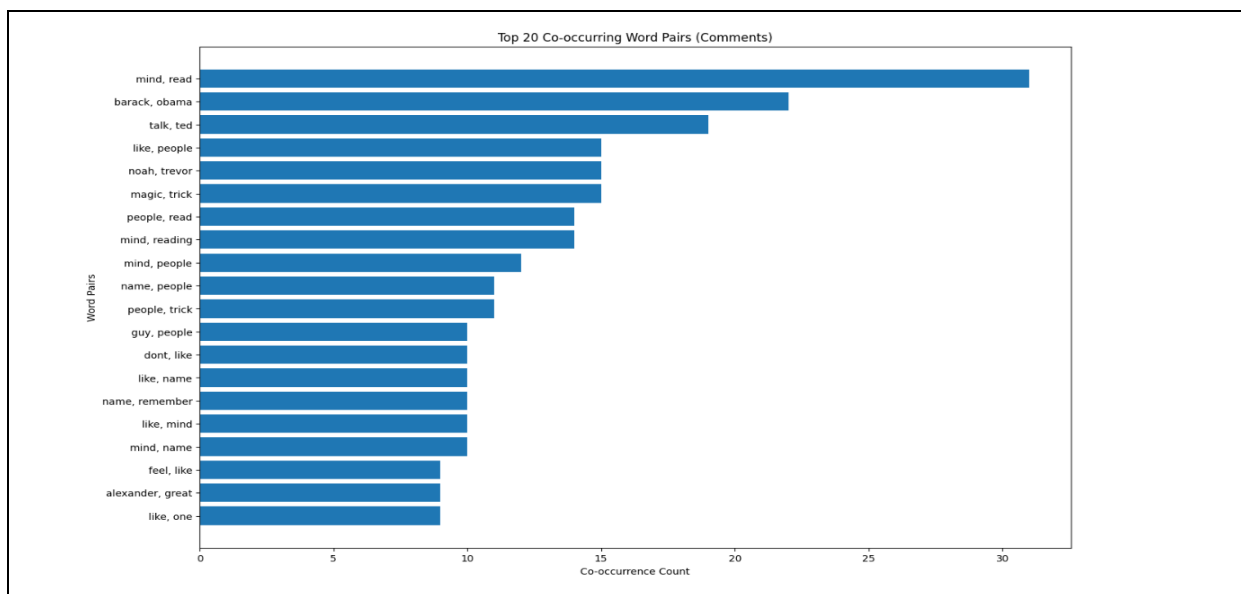
- Comments show a wider variety of frequently used words, reflecting diverse user opinions and expressions.
- Captions contain more repetitive words, which is expected due to scripted or auto-generated caption structures.
- Unigram analysis provides useful baseline information but does not capture relationships between words.

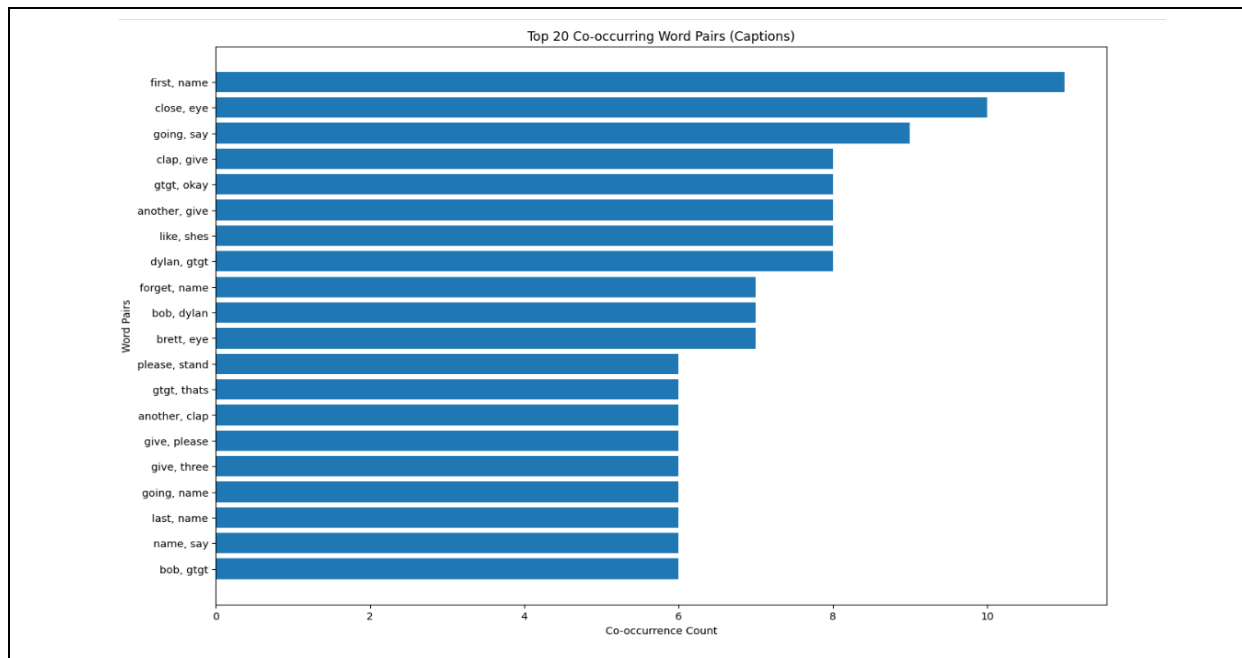
7.2 Frequent Co-occurring Word Pairs

Pairwise co-occurrence analysis was conducted to identify words that frequently appear together within the same transaction.

Results:

- Total unique co-occurring pairs before filtering:
 - Comments: 49,801
 - Captions: 3,448
- After retaining pairs with count ≥ 3 :
 - Comments: 871
 - Captions: 244





Discussion:

- Co-occurrence analysis reveals meaningful associations that unigram frequency alone cannot show.
- Some words appear frequently individually but form strong insights only when analyzed as pairs.
- Comment data produced richer co-occurrence patterns due to longer transaction lengths.

7.3 Co-occurrence Network Graph

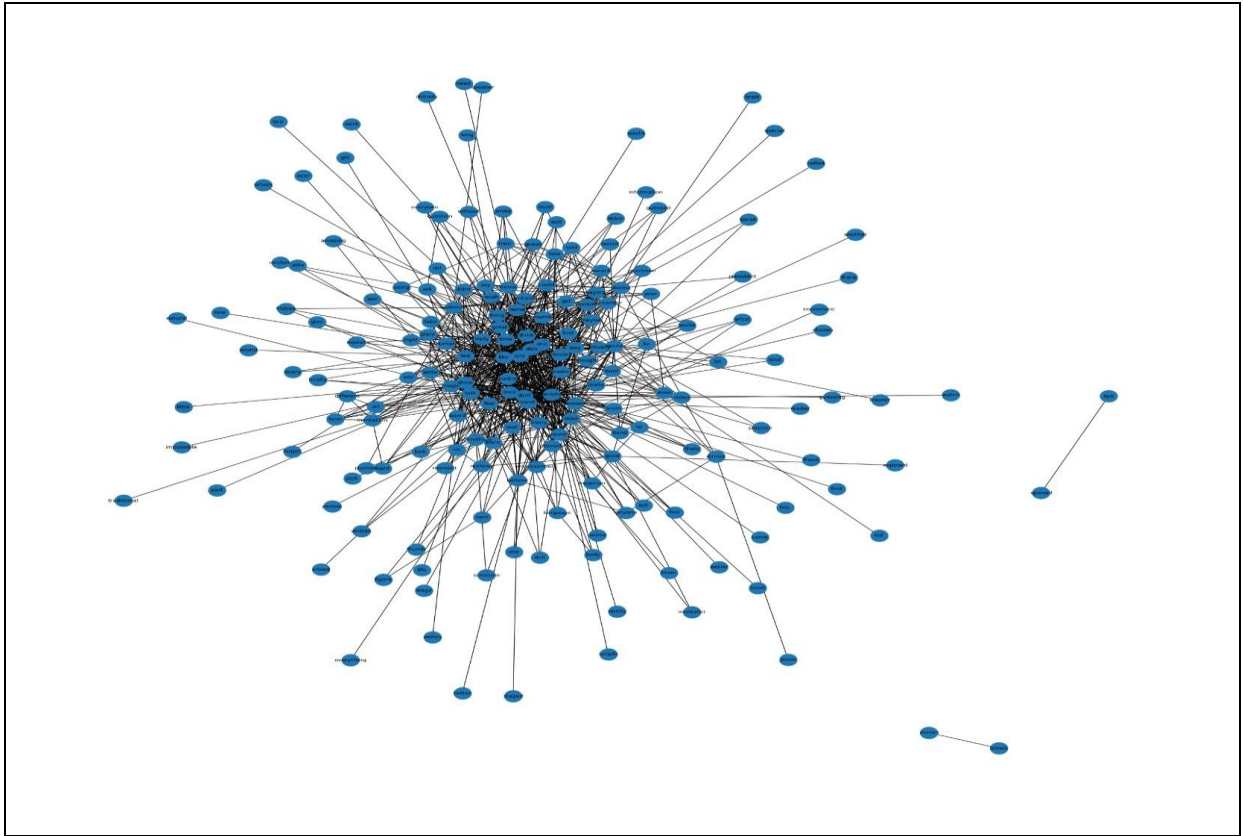
A network graph was constructed using NetworkX to visually represent word relationships.

Results:

- Nodes represent words.
- Edges represent co-occurrence relationships.
- Stopwords were removed to reduce noise.

Discussion:

- Comment networks were larger and denser, indicating complex word relationships.
- Caption networks were smaller with fewer connections, reflecting concise language.
- Clusters in the network highlight groups of semantically related words.



7.5 Comparative Discussion

Aspect	Comments	Captions
Length	Longer	Shorter
Vocabulary Diversity	High	Low
Noise Level	Higher	Lower
Pattern Strength	Weak associations	Limited patterns

7.6 Support vs Confidence Plot Discussion

In association rule mining, a support vs confidence scatter plot is commonly used to visualize the strength and reliability of discovered rules. In this experiment, such plots could not be generated meaningfully due to the absence of strong association rules under the selected thresholds.

7.7 Manual vs Algorithmic Patterns:

- Manual co-occurrence analysis successfully identified frequently appearing word pairs by directly counting word combinations across transactions.
- In contrast, algorithmic pattern discovery (Apriori) relies on strict support and confidence thresholds, which limited the discovery of frequent itemsets.
- This shows that manual co-occurrence methods are more suitable for sparse, short-text datasets, while Apriori is more effective for dense transactional data.

7.8 Impact of Changing min_support:

- Higher support values (0.3, 0.2) resulted in no frequent itemsets.
- Lowering min_support increased the number of itemsets but reduced their statistical strength and generalizability.
- This trade-off highlights the sensitivity of Apriori to parameter selection in text-based datasets.

7.9 Insights from YouTube Data:

- The data reflects diverse and informal user behavior, where words occur frequently on their own but rarely in consistent combinations.
- Emerging topics are better captured through co-occurrence networks and word clouds than through association rules.

7.10 Challenges Faced:

- Data sparsity: Large vocabulary with limited repetition.

- Redundant or weak patterns: Many itemsets appeared only due to low support thresholds.
- Short transaction length: Especially in captions, limiting multi-word associations.

8. Conclusion

This lab explored frequent word pattern discovery and co-occurrence analysis on YouTube comments and captions using both manual and algorithmic approaches.

Key findings include:

- Unigram analysis identified commonly used words but lacked relational insight.
- Manual co-occurrence analysis revealed meaningful word relationships and clusters.
- Apriori-based association rule mining produced limited results due to high lexical diversity and sparse data.
- Comments were richer but noisier, while captions were shorter and more repetitive.
- Visualization techniques provided clearer insights than strict association rules.

Overall, the study demonstrates that co-occurrence-based methods are more effective than Apriori for short, informal text such as YouTube data.

9. Reflection

9.1. Surprising Pairs or Rules

One surprising observation was that some high-frequency words did not form strong co-occurrence pairs. Despite frequent individual appearances, these words rarely occurred together consistently, highlighting the difference between frequency and association.

9.2 Challenging in discovering patterns

The main challenges included:

- High sparsity in one-hot encoded data
- Short and informal text structure
- Difficulty in selecting appropriate support and confidence thresholds
- Lack of strong, repeatable word combinations across transactions

9.3 Future Exploration and Extensions

In future work, the analysis could be extended by:

- Applying sentiment analysis to understand emotional trends in comments
- Using topic modeling (e.g., LDA) to identify hidden themes
- Exploring time-based trends in word usage
- Combining text mining with engagement metrics such as likes or replies